

entry points :

**Pathways and Possibilities to Support Student Learning
about Data and AI**

EVENT #2 | K-12 PROJECTS & OPPORTUNITIES

Sponsored by the College of Education, Health and Human Sciences at
the University of Tennessee, Knoxville

March 31, 2026



COLLEGE OF EDUCATION,
HEALTH & HUMAN SCIENCES

Entry Points Presenters



Data Science 4 Everyone:
Into Data Science & AI Education
Mr. Mahmoud Harding



**University of Tennessee,
Knoxville:**
Dr. Lynn Hodge, Dr. Rachel Wong, &
Dr. Kelly Boles

01

Into Data Science & AI Education

Entry Points

Into Data Science & AI Education

Mahmoud Harding, Instructional Design Director, Data Science 4 Everyone 

Data Science 4 Everyone (DS4E)



Who we are and what we support



is a coalition supporting the advancement of data science education so that every K-12 student is equipped with the data skills needed to succeed in our modern world.

Data Skills

=

Data Literacy

+

Data Science

How does define data literacy?

Data literacy refers to the ability to read, understand, interpret, and communicate data or claims derived from data (Sukol 2025).

How does define data science?

Data science is an interdisciplinary field of study that uses concepts and tools from statistics, computer science, and mathematics to ask and answer questions using data (Sukol 2025).

What is the critical mission of ?

Catalyze the adoption of **both** data literacy and data science as integrated components of K-12 education by 2030.



Why do we support data science education?



Gr. 8 students NAEP mathematics

Content area	2019	2022	Change
Number properties and operations	279		↓
Measurement	280		↓
Data analysis, statistics, & probability	279		↓

Source: Drozda, 2023

College degree and major matching

Students don't always work in jobs aligned to their degrees

The other 2/3 need experience with data skills across disciplines.

✓ **1/3** work in a job that is aligned with their college degree

Source: Federal Reserve Bank of New York, 2013

Data skills are 4 everyone

22.7% of job postings in the US are asking for at least one skill from the category getting, exploring, and analyzing data.



Designed by Freepik

Source: Salerno & Steemers (2024), Burning Glass Institute & ExcelinEd

How can you facilitate data enabled instruction?



3 models of data science instruction

Dedicated Course

A standalone data science course focused on introductory data science skills.

Content Modules

Data literacy & data science are taught through dedicated content modules.

Integration

Data literacy & data science concepts integrated across all grade levels and all subject areas.

All models rely on data-enabled instructional practices.

5 themes of dedicated data science

1. Inquiry-based and iterative investigation
2. Integration of technology
3. Multivariate datasets
4. Mathematical modeling
5. Ethical communication with data



Designed by rawpixel on Freepik

Data Science 4 Everyone

DS4E Content Partners

Evaluation Rubric

Data Science 4 Everyone Content Partner Network

Comprehensive data science education resources across diverse implementation approaches

6

Full Course Offerings

6

Modular & Supplemental

5

Specialized Resources



Full Course Implementations



Find Your Perfect Data Science Tool Match

Filter tools by grade level, time commitment, and coding level to find your classroom fit.

GRADE

Elementary

Middle School

High School

Higher Ed

TIME COMMITMENT

Low (1-2)

Medium (3)

High (4-5)

C

Clear

Apply Filters

Home > Five basic concepts for teachers new to data science

5 basic concepts for teachers new to data science

... you're probably already doing more data science than you realize.

<https://bit.ly/4doZnwt>

Getting Started 5 min read



Integration

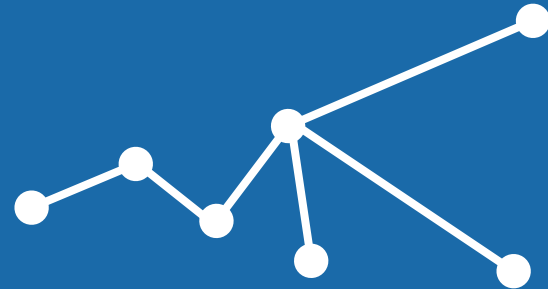
Educator Professional Development: Onboarding to the Data Science Learning Progressions

Hannah Kurzweil

1 What is Data Science Education?



Additional resources for
incorporating data skills
into your instruction &
coursework





datascience4everyone.org/

Substack ds4e.substack.com/

- [There's More to Data Science than Math and Programming](#)
- [Making Data Moves: The Prep Work Behind Every Good Analysis](#)
- [Data Investigation Processes: Connected, Iterative, and Cyclic](#)

DSLPS datasciencelearning.org/

- [Five basic concepts for teachers new to data science](#)

Data articles & resources we recommend

- Wickham, H. (2014). *Tidy Data*. *Journal of Statistical Software*, 59(10), 1–23.
<https://doi.org/10.18637/jss.v059.i10>
- Erickson, T., Wilkerson, M., Finzer, W., & Reichsman, F. (2019). Data Moves. *Technology Innovations in Statistics Education*, 12(1). <http://dx.doi.org/10.5070/T5121038001>
- Kim, A. Y, Ismay, C., & Chunn, J. (2018). The fivethirtyeight R Package: "Tame Data" Principles for Introductory Statistics and Data Science Courses. *Technology Innovations in Statistics Education*, 11(1). <http://dx.doi.org/10.5070/T5111035892>
- Hudson, Rick A., Lee, Hollylynn S., et al. (2024). *Data moves as a focusing lens for learning to teach with data and technology*. *Computers in the Schools*.
<https://doi.org/10.1080/07380569.2024.2411705>
- [Teach Data Literacy: a guide for primary teachers](#)

AI resources we recommend

- Chiu, T. K. F. (2025). AI literacy and competency: definitions, frameworks, development and future research directions. *Interactive Learning Environments*, 33(5), 3225–3229.
<https://doi.org/10.1080/10494820.2025.2514372>
- Modern Language Association of America. (2024). *Student guide to AI literacy*.
<https://d1oemxgau9vshd.cloudfront.net/app/uploads/sites/6/2024/11/AI-Student-Handout-November-2024.pdf>
- [Teaching Responsible AI Literacy in Schools](#)

Sites, papers, articles, etc. mentioned in the presentation

- [DS4E Content Partner Network](#)
- [DS4E Technology Tools for Data Science](#)
- [5 Basic Concepts for Teachers New to Data Science](#)
- [Data Science Starter Kit](#)

02

UTK: Developing Partnerships with Local Communities

Developing Partnerships with Local Communities: An Overview of Three Projects

Dr. Lynn Hodge and
Dr. Rachel Wong



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DEPARTMENT OF
THEORY & PRACTICE IN
TEACHER EDUCATION



Strengthening and Developing Partnerships in East Tennessee for Community Engagement in Artificial Intelligence Education (PARTNER-AI)



THE UNIVERSITY OF
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DEPARTMENT OF
THEORY & PRACTICE IN
TEACHER EDUCATION



This material is based upon work supported by the National Science Foundation under Grant No. 2405213. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the NSF.



Project Background

- Funded by a yearlong U.S. National Science Foundation partnership development grant (Award #2405213)
- **Three project goals:**
 - strengthen and expand existing research-practice partnerships (RPPs) with East Tennessee teachers and school leaders,
 - develop new RPPs with parents and students enrolled in East Tennessee middle and high schools, and
 - co-construct a shared vision for AI that aligns with the needs and assets of the partner community.

PARTNER-AI Project Team

• UTK Researchers

- Rachel Wong (TPTE)
- Lynn Hodge (TPTE)
- Katie Schuman (EECS)
- Joan Williams (TPTE)

• RPP Team

- Megan Reece (Parent)
- Jessica Moore (Parent)
- Patricia Bowling (Parent)
- Student 1
- Student 2
- Jessica Chambers (School Leader – Sevier County)
- Amy Jeffers (School Leader – Scott County)
- Eddie Patton (Teacher – Roane County)
- Joyce Keeton (Teacher – Scott County)

Project Participants

- 19 Teachers
- 4 Parents
- 1 School Leader
- 4 Students

Project Structure

Phase One



UTK Researchers



RPP Team

Phase Two



UTK Researchers



RPP Team

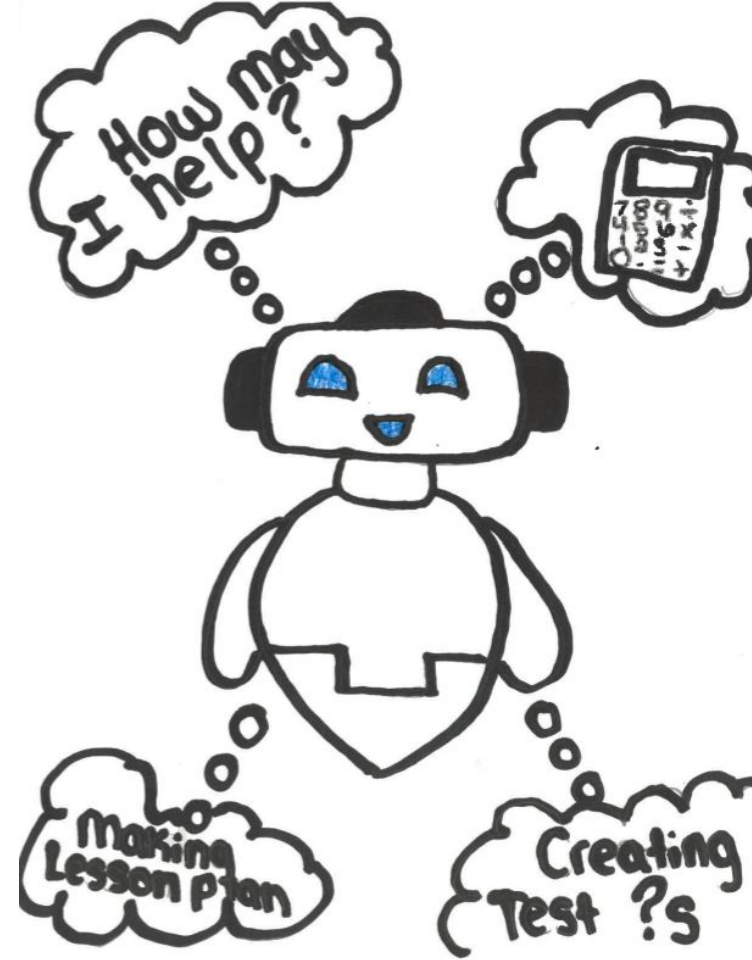


Research
Participants



Phase 1 Activities

- Value Mapping
- Gallery Walk
- Mission and Vision Statements
- Making Sense of Data from Phase 2 Participants



Mission Statement

- Our mission is to strengthen educational experiences across East Tennessee by using AI as a supportive tool for teaching and learning. We are committed to cultivating collaborative learning communities—whether virtual or in person—where students and teachers can meaningfully engage with emerging technologies. By prioritizing critical thinking, transparency, and ethical use of AI, we aim to equip all learners with the skills to analyze, evaluate, and apply AI responsibly and effectively.

Vision Statement

- To thoughtfully integrate AI to enhance education, uphold ethical and responsible practices, and inspire innovation. We aim to empower students and educators by equipping them with the skills to think critically, collaborate effectively, solve real-world challenges, and use AI thoughtfully and responsibly.

Phase 2 Activities

- Initial Survey
- Interview
- Focus Group
- Final Survey

Common Themes Across All Activities

Concerns about
Over-Reliance

Need for Clear
Policies,
Boundaries, and
Training

AI as a Tool for
Efficiency and
Support

Concerns about
Erosion of
Critical Thinking

Mixed
Feelings/Views
on AI



Next Steps

- Themes point to the need for intentional training and support:
 - Strengthen and further our work with Roane County Schools to co-design and implement AI PD curriculum for teachers that directly address some of these concerns

MATHEMATIZING, VISUALIZING, AND POWER (MVP): APPALACHIAN YOUTH BECOMING DATA ARTISTS FOR COMMUNITY LEARNING

I

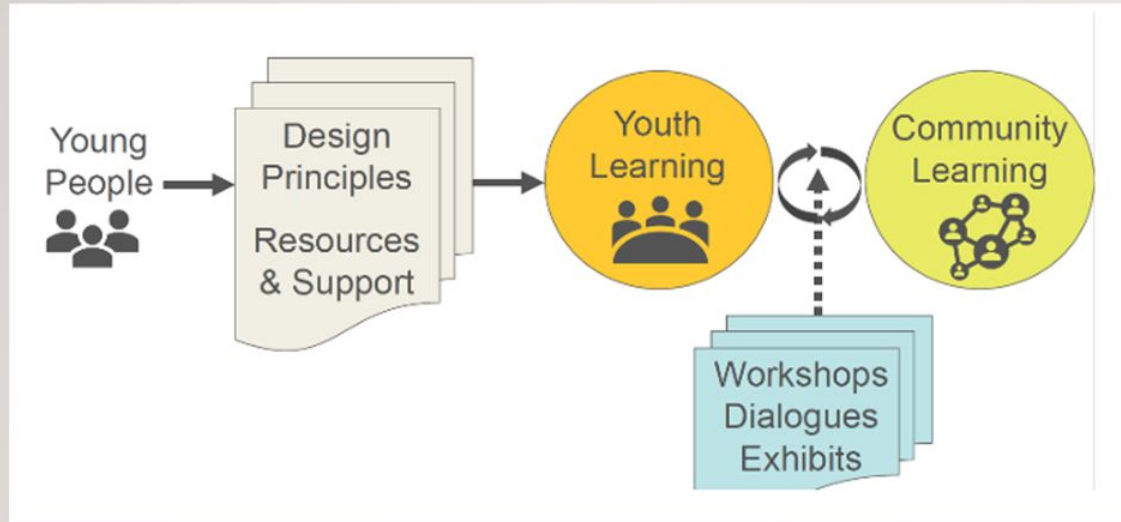
MVP PROJECT TEAM

- Joy Bertling (UTK)
- Rhea Carmon (UTK)
- Chulin Chen (UTK)
- Carlye Clark (Boys and Girls Club)
- Caroline Covington (Pellissippi State Community College - PSCC)
- Mary Katherine Dailey (Knox County Schools)
- Lauren Farkas (Knox County Schools)
- Kristina Givens (Knox County Schools)
- Carlos Gonzalez (UTK)
- Rio Gonzalez (PSCC)
- Amy Maples (UTK)
- Megan Mundie (UTK)
- Kylee Pace (Knox County Schools)
- Ethan Pignataro (Knox County Schools)
- Lee Reinert (Knox County Schools)
- Matt Riddle (UTK)
- George Schafer (Drexel)
- Jessica Schwind (Knox County Schools)
- Ashley Stroud (UTK)
- Trey Smith (Philadelphia Public Schools & U. of Pennsylvania)
- Christopher Wright (Drexel)
- Yilang Zhao (University of Buffalo)

MVP PROJECT BACKGROUND

- Funded by a three-year U. S. National Science Foundation grant (Award #2215004); PI Team: Hodge, Dyer, Bertling, & Clark
- Overarching purpose: Test, implement, and study a model for community data engagement catalyzed by youth and their self-created data visualizations
- Theoretical framing: Social and Cultural perspectives with an emphasis on critical and community-centered approaches to working with data (D'Ignazio & Klein, 2020; Philip et al., 2016; Rubel et al., 2021); hybridity (Gutierrez et al., 1999), and youth agency (Moses & Cobb, 2021)
- Currently in year 4 (extension year)

MVP MODEL AND CONTEXT



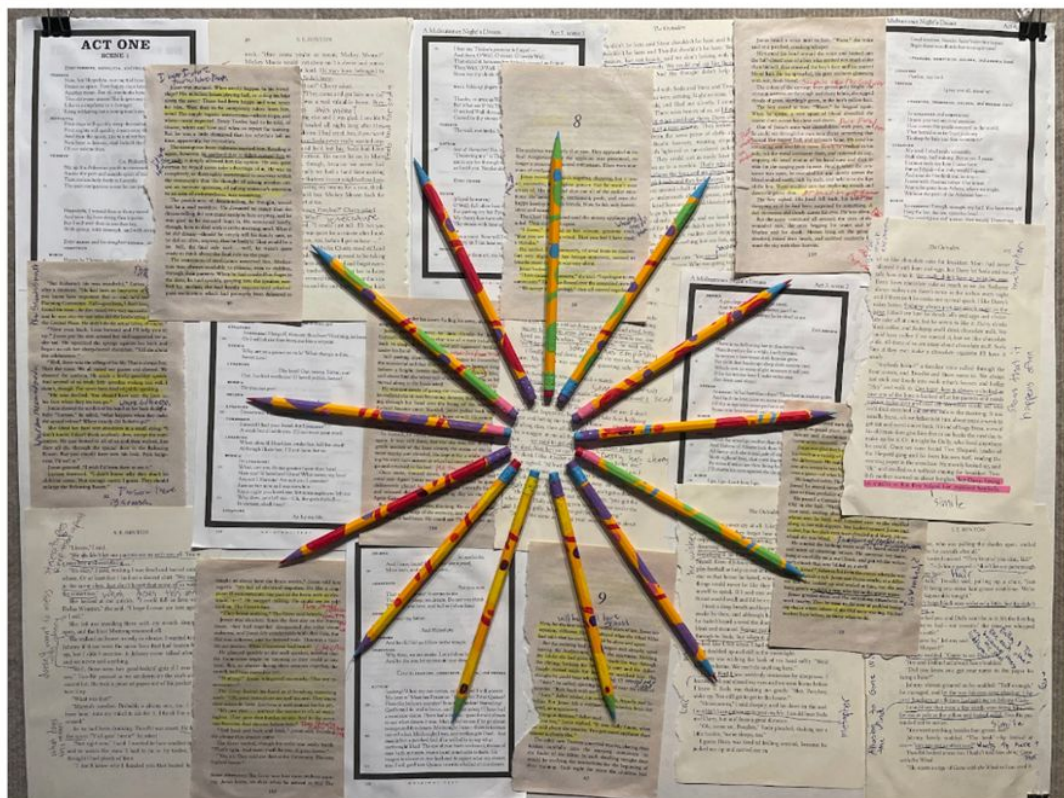
Timing: Cycles 1-4, Spring 2023 – Summer 2024

Youth Learning:

82 youth participants
– known as data artists –
ages 12-17) across 4
cohorts participated in
weekly sessions

Community Learning

1-3 events per cycle



In Our Words Paint, paper, pencil

“My piece shows how my friends and I feel about middle school and high school. The pages are words that we read through middle school, while the paint and pencils represent the emotions we feel.”

-Karen, grade 8



Parental Styles and Relationships
Paint, marker

“This artwork represents the responses gathered from those primarily aged 12-16 about whether or not their parents are strict or whether they get along with said parents.”

-Nina, grade 11



Losing Sleep

Embroidery hoop, suede leather lace, hemp cord, and beads

“...My piece is about sleep schedules and the impact school/work has on them. My data is visualized in a dream catcher to represent hopes for peaceful nights of sleep. The beads display the data with each bead type representing a different answer choice from my survey. Each string is a person who took the survey...”

-Dakota, Grade 12

CURRENT WORK

- Planning for a community Workshop in May to revisit the curriculum and facilitators guide
- Research:
 - What are data artists identities in relation to community as reflected in their data visualizations?
 - How is the integration of data and art reflected in the student-created visualizations? How do the data visualizations sustain viewers' interests?
- <https://mvp-project.com>

AI Presidential Challenge



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TEACHER EDUCATION



AI Presidential Challenge Project

- Led by Dr. Yukyeong Song
- Mentors included Dr. Rachel Wong, Dr. Vinhthuy Phan (UM), and Dr. Jewoong Moon (UA)
- National challenge where educators, mentors, K-12 students, and community teams come together to solve real-world problems in their community using AI-Powered solutions.

AI Presidential Challenge

- Three K-12 teachers from Roane County Schools
- 8-week-long weekly online structured workshop
- Design thinking framework
- Math Mysteries

Welcome to Math Mysteries

A narrative-driven math learning platform for middle schoolers that builds confidence through storytelling.

Math isn't just numbers—it's the key to solving the case. Dive into immersive stories, use your detective notebook, and apply logic to catch the culprit.

No pressure. No timers. Just you and the mystery.

Select Your Grade Level

6th Grade

7th Grade

8th Grade

Select your grade to filter mysteries by the standards you are learning.

Choose Your Investigation

Select a case file below to begin. Each mission contains a unique story generated just for you.



NOVICE

The Ratio Recipe

Solve a culinary mystery using ratios and rates.

TOPIC: TN STANDARD 6.RP.A.3: USE RATIO AND RATE REASONING TO SOLVE REAL-WORLD AND MATHEMATICAL PROBLEMS.

Goal: TN Standard 6.RP.A.3 — Master equivalent ratios and unit rates.

Understanding ratios is essential for adjusting recipes, converting currencies, and analyzing relationships between quantities in daily life.



APPRENTICE

The Data Detective

Analyze the clues found at the crime scene.

TOPIC: TN STANDARD 6.SP.B.5: SUMMARIZE NUMERICAL DATA SETS IN RELATION TO THEIR CONTEXT.

Goal: TN Standard 6.SP.B.5 — Calculate mean, median, and mode to find the culprit.

Statistical analysis helps us interpret complex information, spot trends, and make evidence-based decisions in fields ranging from sports to medicine.



MASTER

The Negative Number Heist

Track the thief through underground levels.

TOPIC: TN STANDARD 6.NS.C.5: UNDERSTAND THAT POSITIVE AND NEGATIVE NUMBERS ARE USED TOGETHER TO DESCRIBE QUANTITIES.

Goal: TN Standard 6.NS.C.5 — Understand positive and negative integers in real-world contexts.

Integers help us describe temperature, elevation, and financial debts or credits.



EXPERT

The Geometry Giant

Find the area of the lost city districts.

TOPIC: TN STANDARD 6.G.A.1: FIND THE AREA OF RIGHT TRIANGLES, OTHER TRIANGLES, SPECIAL QUADRILATERALS, AND POLYGONS.

Goal: TN Standard 6.G.A.1 — Decompose shapes to find total area.

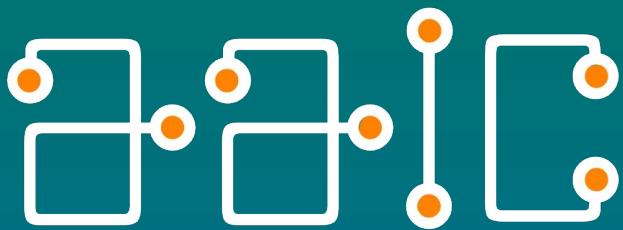
Area calculation is crucial for architecture, land surveying, and interior design.



THEORY & PRACTICE IN
TEACHER EDUCATION

03

**UTK: Data Science
and AI Education
within
Community
Contexts**



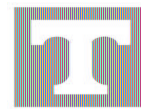
Appalachian AI Corps

Empowering Appalachia Through Artificial Intelligence



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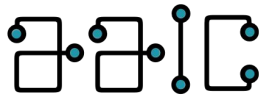
DEPARTMENT OF THEORY &
PRACTICE IN TEACHER EDUCATION



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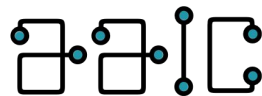
MIN H. KAO DEPARTMENT
ELECTRICAL ENGINEERING
COMPUTER SCIENCE

Students who have opportunities to deeply **learn** about, **use**, and **build with AI tools** in **authentic community contexts**



Students who have opportunities to deeply **learn** about, **use**, and **build with AI tools** in **authentic community contexts**

are students who will be **empowered to make informed decisions** about AI use

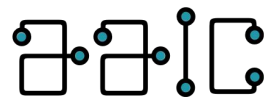


Students who have opportunities to deeply **learn** about, **use**, and **build with AI tools** in **authentic community contexts**

are students who will be **empowered to make informed decisions** about AI use

and

positioned to leverage it for the good of their communities and families.

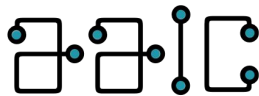


CURRENT WORK:

Water Quality Monitoring

Project Context

- Pilot year 1
- Partnership with a local high school:
 - CTE: Agricultural Science (Agriscience)
 - Environmental Science
- Co-designed six workshops around curricular standards and local community contexts
- Will hold workshop #5 next week



Curricular Alignment



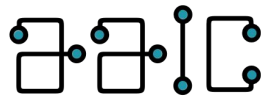
Department of
Education

College, Career and
Technical Education

Agriscience

Supervised Agricultural Experience

- Student-driven
- Should implement multiple science and engineering practices:



Curricular Alignment



Department of
Education

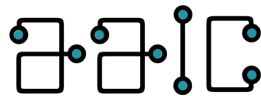
College, Career and
Technical Education

Agriscience

Supervised Agricultural Experience

- Student-driven
- Should implement multiple science and engineering practices:

- a. AQDP - asking questions and developing problems
- b. MOD - developing and using models
- c. PCI - planning and carrying out investigations
- d. AID - analyzing and interpreting data
- e. UMCT - using mathematics and computational thinking
- f. CEDS - constructing explanations (for science) and designing solutions (for engineering)
- g. EAE - engaging in argument from evidence
- h. OEI - obtaining, evaluating, and communicating information



SMOKEY BUOY

As Entry Point

Smokey Buoy



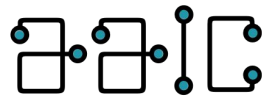
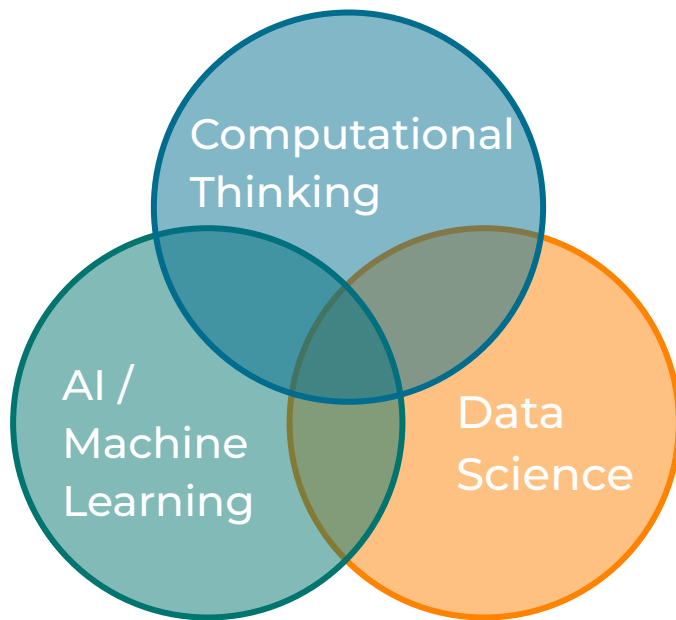
THE UNIVERSITY OF
TENNESSEE

Min H. Kao Department of Electrical Engineering and
Computer Science

Department of Theory & Practice in Teacher Education

THE WORKSHOPS

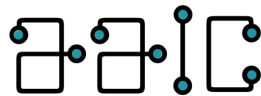
The Workshops: Technical Content



The Workshops: Materials

Each workshop has a robust set of curricular materials, including:

- Slide deck
- Accompanying slide deck with facilitator annotations
- Facilitator overview video
- Facilitator guide with standards alignment, preparation tips, and additional resources
- Student handout and (optional) notetaker
- Online student module with interactive Python code blocks



The Workshops: Topics

W1

Intro to Water Quality & Statistics Fundamentals

W2

Intro to Convolutional Neural Nets (CNNs) & Classification

W3

CNNs & Regression

W4

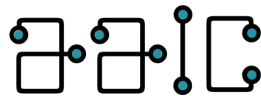
Physical Computing (Arduinos, Motors, Sensors, etc.)

W5

Asking Questions of & Analyzing Buoy Data

W6

Presentation of Findings & Career Connections



The Workshops: Topics

W1

Intro to Water Quality & Statistics Fundamentals

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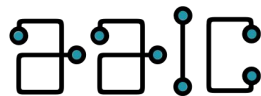
W5

Asking Questions of & Analyzing Buoy Data

W6

Presentation of Findings & Career Connections

**Buoy
Deployment**

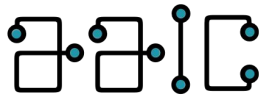


Buoy Deployment



What's Next

- Pilot Year 1 completed by Summer 2026
- Workshop materials will be refined and released for public use
- Plans to expand to more regional schools/districts next year
 - Contingent on additional funding



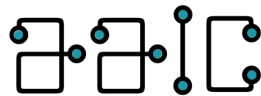
This work is funded through an AI Tennessee seed grant.

It would be impossible to do this work without our UTK team!

Dr. Sukanya Moudgalya
Dr. Kelly L. Boles
Dr. Sai Swaminathan
Dr. Michela Taufer
Dr. John S. Schwartz

Dr. Ian M. Simpson
Dr. Amanda Zeller
Breanna Slessman
Ronald Rupard
Jacob Foltz

**And a special thank you to our wonderful school,
teacher, and student partners!**





COMMUNITY DATA LIBRARIES

Data *about* the
community *for* the
community.



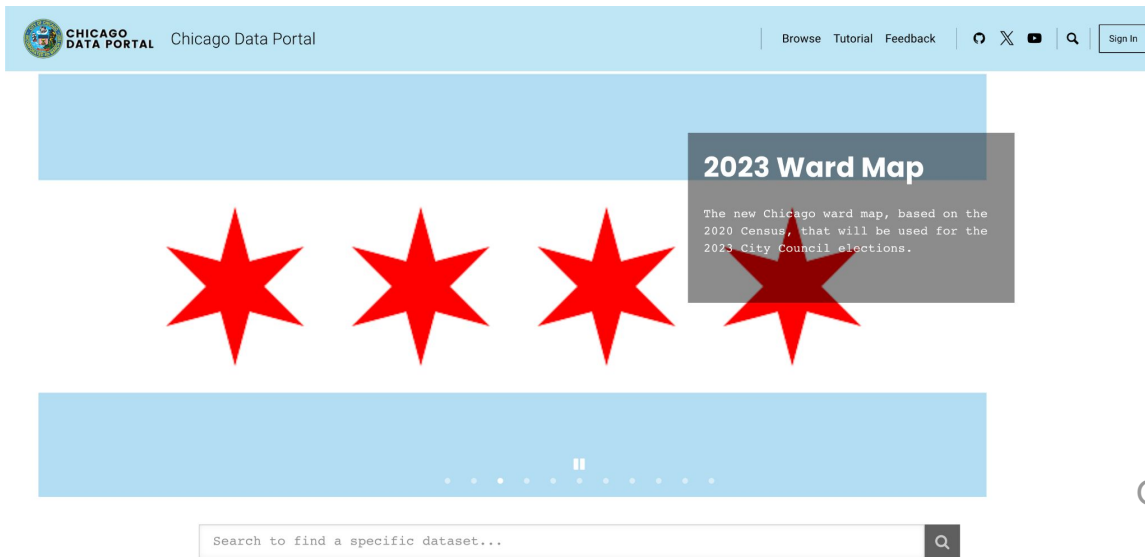
Motivation

- In data science education, it is recommended that data used in instructional activities be contextually-relevant and proximate to the learner
 - Can be achieved in different ways and to different degrees
 - Did the learner generate/create the data set?
 - Is the data about the learner?

(Lee & Delaney, 2022; Weiland & Williams, 2024)

Motivation

- Large cities often have open data repositories for citizens
 - Makes finding data a bit easier, but...
 - Expensive, must be maintained
 - Difficult/impossible for small town and communities





The Project: Community Data Libraries

- Under active development
- Community data libraries will provide a selection of open data sets that contain data about a given community
 - Include features making dataset selection and use easier for particular pedagogical purposes/curricular standards
- An interested community member will be able to create their own local data library using our tools



The Project: Timeline

Phase One

**Active development of
Community Data
Libraries**

Phase Two

**Creation of curricular
materials to integrate
CDLs across content
areas.**

Phase Three

**Teacher professional
learning opportunities
and training around CDL
data and resources**

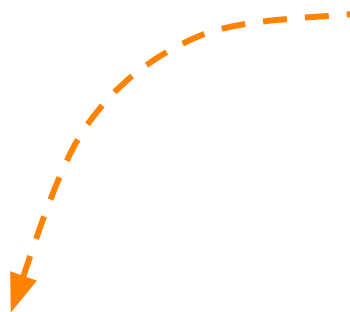
Phase Four

**Community member
workshops using CDL
data and resources.**



The Project: Timeline

Soon: Searching for schools and teachers interested in using CDLs.



Phase One

Active development of Community Data Libraries

Phase Two

Creation of curricular materials to integrate CDLs across content areas.

Phase Three

Teacher professional learning opportunities and training around CDL data and resources

Phase Four

Community member workshops using CDL data and resources.



Acknowledgements

This project is made possible by the support of the BelleJAR Foundation.

Thank you to the CDL team!

Dr. Kelly Boles

Aadya Sharma

Molly McKenzie

Ariel Thompson

Panel Discussion

Q & A



entry points :

**Pathways and Possibilities to Support Student Learning
about Data and AI**

EVENT #3 | NATIONAL FRAMEWORKS & RECOMMENDATIONS (APR 28)

- Data Science Learning Progressions
- National Academies' Competencies for the Future of Data and Computing
- 2025 GAISE College Report Revision (to include Data Science)

entrypointsutk.org



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HEALTH & HUMAN SCIENCES